

Computers Around Us



All around us, technology is growing at a fast pace. It is important for us to stay up to date with its advancements.

We are all familiar with usual computing devices which we use every day. For example: computers and mobile phones. But the computing technology is not limited to these devices.

1. Computers Around Us

Let us look at a few new age technological devices:

Wearables: This type of device can be worn by the user. Wearables are light weight and easy to carry around. These devices are the most popular new age gadgets.

Bytes

Gadgets: Gadgets are mechanical or electronic devices which have practical uses. They are usually invented to fulfil a need. For example: smart phones.

Types of wearables:

Smartwatch

Smartwatch: A smartwatch is a digital device which includes all the features of a smartphone. It is compact enough to fit on our wrists.



Smartwatch

These are some functions a smart watch can perform:

- a. Receive and make calls
- b. Read and reply to messages
- c. Check social media notifications
- d. Set alarms
- e. Use maps to find a location
- f. Count the number of steps taken in a day
- g. Measure heart rate
- h. Listen to and turn off alarms

Similar to traditional computing devices, a smartwatch runs on an operating system. This OS is specifically designed for it.

Fitness Band

Fitness Band: A fitness band is used to track a person's physical activity. It is worn like a watch and uses sensors to track the activity.



Fitness Band

Bytes

Steve Mann designed the first smartwatch in 1998. He is known as the father of wearable computing.

1. Computers Around Us

A fitness band can be used to track the following:



a. Sleep tracker: calculates the number of hours the user has slept for.



b. Step counter: counts the steps taken each day.



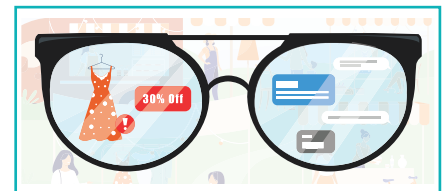
c. Heart rate tracker: tracks heartbeats per minute.

Smart watches and fitness bands look similar. But they are very different from each other:

- Fitness bands do not have memory to run applications like messages, calls or social media.
- Fitness bands have smaller screen sizes.
- Fitness bands may or may not have a touchscreen.

Smart Glasses

Smart Glasses: Smart glasses are eyeglasses that provide information to the wearer about what he sees. For example: if the wearer is looking at a car, the smart glasses would provide information about that car. They have built in functions to wirelessly connect to the internet.



Smart Glasses

Smart glasses can be used to:

- a. Search for a location's description
- b. Translate a foreign language
- c. Identify products and their reviews
- d. Take calls and communicate through messages. Artificial intelligence or robotic voice assistants are used for this.

Smart TV

Smart TV: A Smart TV is connected to the internet via Wi-Fi or a broadband connection. It can play videos, movies, and animations available on the internet.



Smart TV

Some common functions of a Smart TV are:

- a. Stream music, videos and radio.
- b. Browse the internet directly on the TV screen.
- c. Access streaming services like Netflix, HULU, and YouTube.
- d. Recognise gestures and voices. This is useful while playing games or searching for information.
- e. Connect to cloud services to browse files, view pictures and play videos online.
- f. Connect to other computing devices via physical ports like HDMI cables. They can also be connected wirelessly via Wi-Fi. For example: A smart TV can be connected to a laptop via Wi-Fi.

Exercise

Fill in the blanks

- 1 _____ do not run applications like smartwatches.
- 2 _____ helps us identify products and find reviews.

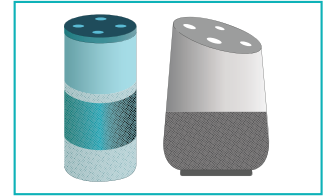
State whether True or False

- 3 Fitness bands track a user's activity.
 _____
- 4 A Smart TV can be used to search the web.
 _____
- 5 Smart Watches come packaged with virtual assistants.
 _____

1. Computers Around Us

Smart speakers

Smart speakers: A speaker is used to listen to audio. It connects to another device like a smartphone or computer. A smart speaker functions through voice commands. It is also called a Home Assistant.



Speakers

- a. The user can talk to a virtual assistant and ask it to perform tasks.
- b. It can perform operations to play music on request, search the web, tell us the news and weather, and answer questions.
- c. A smart speaker must be connected to the internet to perform any of these tasks.
- d. It does not contain a CPU.

Futuristic Cars

Traditionally, cars were only used to go from one place to another. But in the digital age, vehicles have been transformed to perform other functions as well.



Smart Car

Some common functions of modern-day cars are:

- a. GPS navigation: It uses satellites to show the destination route.
- b. Wireless connectivity: Connects to the internet to play music and videos.
- c. Voice recognition and control: We can use voice commands to give instructions.
- d. Artificial Intelligence enabled proximity sensors: They detect closeness to nearby objects. These sensors are used in cars to avoid crashes and prevent accidents.
- e. AI operated self-driving cars (Autopilot feature): Autopilot is a mode in self-driving cars. The computer/AI in the car takes control of the steering.
- f. Built in security cameras connected to Wi-Fi.
- g. Automatic access to a smartphone or tablet.



Futuristic cars have a lot of other features. These are the most common ones. As technology progresses, most cars will include them.

In this lesson, we have learnt about the leaps that technology has made in the 21st century. Computing devices have advanced from computers to self-driving cars. It will be interesting to see what technology brings us next!

Exercise

Fill in the blanks

- 6 Smart speakers are sometimes called _____.
- 7 _____ help futuristic cars avoid collisions.
- 8 _____ cars are controlled by Artificial intelligence. This mode is also called autopilot.

Match the following

- | | |
|--------------------------|--------------------|
| 9 a. Autopilot | f. Smart TV |
| b. Watch OS | g. Smart Glasses |
| c. Heart Rate Monitoring | h. Smart Watch |
| d. Video Streaming | i. Fitness Band |
| e. Translate Languages | j. Futuristic Cars |



Tech Flash

- **Wearables** : Digital devices that can be worn by the user are called wearables.
- **Smart Watch** : Digital watches which have basic computational features as well as an operating system are called smart watches.
- **Smart TV** : Smart TVs are TVs which can be connected to the internet. They run on an operating system meant for TVs.
- **Smart Speakers** : A speaker which is packaged with a personalized virtual assistant is a smart speaker. It is voice controlled and uses the internet to perform tasks like music streaming and web search.
- **Autopilot** : Autopilot is a mode in self driving cars where the computer/AI in the car takes control of the steering.
- **Proximity Sensor** : This is a sensor which detects closeness to nearby objects. It is used in cars to avoid collisions.